

Assignment 1

Due at 11:59pm on September 16.

Zupeng Zeng

This assignment is to be submitted individually. Turn in this assignment as an HTML or PDF file to ELMS. Make sure to include the R Markdown or Quarto file that was used to generate it. You should include the questions in your solutions. You may use the qmd file of the assignment provided to insert your answers.

Git and GitHub

1) Provide the link to the GitHub repo that you used to practice git from Week 1. It should have:

- Your name on the README file.
- At least one commit with your name, with a description of what you did in that commit.

<https://github.com/zzeng05/Zeng1-Liu2-a1.git>

Reading Data

Download both the Angell.dta (Stata data format) dataset and the Angell.txt dataset from this website: <https://stats.idre.ucla.edu/stata/examples/ara/applied-regression-analysis-by-fox-data-files/>

2) Read in the .dta version and store in an object called `angell_stata`.

```
library(haven)
angell_stata <- read_dta("/Users/zpzzz/Desktop/SURV727/Zeng1-Liu2-a1/angell.dta")
summary(angell_stata)
```

city	morint	ethhet	geomob
Length:43	Min. : 4.20	Min. :10.60	Min. :12.10
Class :character	1st Qu.: 8.70	1st Qu.:16.90	1st Qu.:19.45
Mode :character	Median :11.10	Median :23.70	Median :25.90
	Mean :11.20	Mean :31.37	Mean :27.60
	3rd Qu.:13.95	3rd Qu.:39.00	3rd Qu.:34.80
	Max. :19.00	Max. :84.50	Max. :49.80

region
Length:43
Class :character
Mode :character

3) Read in the .txt version and store it in an object called `angell_txt`.

```
angell_txt <- read.table("/Users/zpzzz/Desktop/SURV727/Zeng1-Liu2-a1/angell.txt")
summary(angell_txt)
```

V1	V2	V3	V4
Length:43	Min. : 4.20	Min. :10.60	Min. :12.10
Class :character	1st Qu.: 8.70	1st Qu.:16.90	1st Qu.:19.45
Mode :character	Median :11.10	Median :23.70	Median :25.90
	Mean :11.20	Mean :31.37	Mean :27.60
	3rd Qu.:13.95	3rd Qu.:39.00	3rd Qu.:34.80
	Max. :19.00	Max. :84.50	Max. :49.80

V5
Length:43
Class :character
Mode :character

4) What are the differences between `angell_stata` and `angell_txt`? Are there differences in the classes of the individual columns?

```
names(angell_stata)
```

```
[1] "city" "morint" "ethhet" "geomob" "region"
```

```
names(angell_txt)
```

```
[1] "V1" "V2" "V3" "V4" "V5"
```

```
sapply(angell_stata, class)
```

city	morint	ethhet	geomob	region
"character"	"numeric"	"numeric"	"numeric"	"character"

```
sapply(angell_txt, class)
```

V1	V2	V3	V4	V5
"character"	"numeric"	"numeric"	"numeric"	"character"

```
head(angell_txt)
```

	V1	V2	V3	V4	V5
1	Rochester	19.0	20.6	15.0	E
2	Syracuse	17.0	15.6	20.2	E
3	Worcester	16.4	22.1	13.6	E
4	Erie	16.2	14.0	14.8	E
5	Milwaukee	15.8	17.4	17.6	MW
6	Bridgeport	15.3	27.9	17.5	E

```
head(angell_stata)
```

```
# A tibble: 6 x 5
  city      morint ethhet geomob region
  <chr>    <dbl>  <dbl>  <dbl> <chr>
1 Rochester    19    20.6    15    E
2 Syracuse    17    15.6   20.2    E
3 Worcester   16.4    22.1   13.6    E
4 Erie        16.2    14     14.8    E
5 Milwaukee   15.8    17.4   17.6    MW
6 Bridgeport   15.3    27.9   17.5    E
```

5) Make any updates necessary so that `angell_txt` is the same as `angell_stata`.

```
## Rename columns
names(angell_txt) <- names(angell_stata)
```

```
## Cross check
print(names(angell_stata))
```

```
[1] "city"    "morint" "ethhet" "geomob" "region"
```

```
print(names(angell_txt))
```

```
[1] "city"    "morint" "ethhet" "geomob" "region"
```

```
print(head(angell_stata))
```

```
# A tibble: 6 x 5
  city      morint ethhet geomob region
<chr>    <dbl>  <dbl>  <dbl> <chr>
1 Rochester    19    20.6    15    E
2 Syracuse    17    15.6   20.2    E
3 Worcester   16.4   22.1   13.6    E
4 Erie        16.2    14    14.8    E
5 Milwaukee   15.8   17.4   17.6    MW
6 Bridgeport  15.3   27.9   17.5    E
```

```
print(head(angell_txt))
```

```
      city morint ethhet geomob region
1 Rochester  19.0  20.6  15.0      E
2  Syracuse  17.0  15.6  20.2      E
3 Worcester  16.4  22.1  13.6      E
4      Erie  16.2  14.0  14.8      E
5 Milwaukee  15.8  17.4  17.6      MW
6 Bridgeport  15.3  27.9  17.5      E
```

```
print(all.equal(as.data.frame(angell_txt),
                  as.data.frame(angell_stata),
                  check.attributes = FALSE))
```

```
[1] "Component \"morint\": Mean relative difference: 2.311301e-08"
[2] "Component \"ethhet\": Mean relative difference: 2.475718e-08"
[3] "Component \"geomob\": Mean relative difference: 2.394829e-08"
```

6) Describe the Ethnic Heterogeneity variable. Use descriptive statistics such as mean, median, standard deviation, etc. How does it differ by region?

```
library(psych)

psych::describe(angell_stata$ethhet)
```

```
      vars  n  mean    sd median trimmed   mad  min  max range skew kurtosis   se
X1      1 43 31.37 20.41   23.7   28.33 12.01 10.6 84.5  73.9 1.18      0.3 3.11
```

```
psych::describeBy(angell_stata$ethhet, angell_stata$region, mat = TRUE)
```

```
      item group1 vars  n    mean      sd median  trimmed      mad  min  max
X11     1      E   1  9 23.48889 10.773398  22.10 23.48889  9.636900 10.6 45.8
X12     2     MW   1 14 21.67857  9.084914  19.25 21.09167  8.154301 10.7 39.7
X13     3      S   1 14 52.48571 21.440897  53.80 52.46667 27.353968 20.7 84.5
X14     4      W   1  6 16.55000  4.164012  16.15 16.55000  3.558239 12.3 23.9
      range      skew  kurtosis      se
X11  35.2  0.747188465 -0.5716749 3.591133
X12  29.0  0.734178773 -0.7125926 2.428045
X13  63.8 -0.001023661 -1.4947575 5.730321
X14  11.6  0.639974099 -1.1117407 1.699951
```

Describing Data

R comes also with many built-in datasets. The “MASS” package, for example, comes with the “Boston” dataset.

7) Install the “MASS” package, load the package. Then, load the Boston dataset.

```
if (!requireNamespace("MASS", quietly = TRUE)) install.packages("MASS") # for smooth rendering
library(MASS)
data("Boston")
```

8) What is the type of the Boston object?

```
typeof(Boston)
```

```
[1] "list"
```

9) What is the class of the Boston object?

```
class(Boston)
```

```
[1] "data.frame"
```

10) How many of the suburbs in the Boston data set bound the Charles river?

```
head(Boston)
```

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	black	lstat
1	0.00632	18	2.31	0	0.538	6.575	65.2	4.0900	1	296	15.3	396.90	4.98
2	0.02731	0	7.07	0	0.469	6.421	78.9	4.9671	2	242	17.8	396.90	9.14
3	0.02729	0	7.07	0	0.469	7.185	61.1	4.9671	2	242	17.8	392.83	4.03
4	0.03237	0	2.18	0	0.458	6.998	45.8	6.0622	3	222	18.7	394.63	2.94
5	0.06905	0	2.18	0	0.458	7.147	54.2	6.0622	3	222	18.7	396.90	5.33
6	0.02985	0	2.18	0	0.458	6.430	58.7	6.0622	3	222	18.7	394.12	5.21

	medv
1	24.0
2	21.6
3	34.7
4	33.4
5	36.2
6	28.7

```
sum(Boston$chas == 1)
```

```
[1] 35
```

11) Do any of the suburbs of Boston appear to have particularly high crime rates? Tax rates? Pupil-teacher ratios? Comment on the range of each variable.

```
head(Boston[order(-Boston$crim), ])
```

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	black	lstat
381	88.9762	0	18.1	0	0.671	6.968	91.9	1.4165	24	666	20.2	396.90	17.21
419	73.5341	0	18.1	0	0.679	5.957	100.0	1.8026	24	666	20.2	16.45	20.62
406	67.9208	0	18.1	0	0.693	5.683	100.0	1.4254	24	666	20.2	384.97	22.98
411	51.1358	0	18.1	0	0.597	5.757	100.0	1.4130	24	666	20.2	2.60	10.11
415	45.7461	0	18.1	0	0.693	4.519	100.0	1.6582	24	666	20.2	88.27	36.98
405	41.5292	0	18.1	0	0.693	5.531	85.4	1.6074	24	666	20.2	329.46	27.38
medv													
381	10.4												
419	8.8												
406	5.0												
411	15.0												
415	7.0												
405	8.5												

```
head(Boston[order(-Boston$tax), ])
```

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	black	lstat
489	0.15086	0	27.74	0	0.609	5.454	92.7	1.8209	4	711	20.1	395.09	18.06
490	0.18337	0	27.74	0	0.609	5.414	98.3	1.7554	4	711	20.1	344.05	23.97
491	0.20746	0	27.74	0	0.609	5.093	98.0	1.8226	4	711	20.1	318.43	29.68
492	0.10574	0	27.74	0	0.609	5.983	98.8	1.8681	4	711	20.1	390.11	18.07
493	0.11132	0	27.74	0	0.609	5.983	83.5	2.1099	4	711	20.1	396.90	13.35
357	8.98296	0	18.10	1	0.770	6.212	97.4	2.1222	24	666	20.2	377.73	17.60
medv													
489	15.2												
490	7.0												
491	8.1												
492	13.6												
493	20.1												
357	17.8												

```
head(Boston[order(-Boston$ptratio), ])
```

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	black	lstat
355	0.04301	80	1.91	0	0.413	5.663	21.9	10.5857	4	334	22.0	382.80	8.05
356	0.10659	80	1.91	0	0.413	5.936	19.5	10.5857	4	334	22.0	376.04	5.57
128	0.25915	0	21.89	0	0.624	5.693	96.0	1.7883	4	437	21.2	392.11	17.19
129	0.32543	0	21.89	0	0.624	6.431	98.8	1.8125	4	437	21.2	396.90	15.39
130	0.88125	0	21.89	0	0.624	5.637	94.7	1.9799	4	437	21.2	396.90	18.34
131	0.34006	0	21.89	0	0.624	6.458	98.9	2.1185	4	437	21.2	395.04	12.60

```

      medv
355 18.2
356 20.6
128 16.2
129 18.0
130 14.3
131 19.2

```

```
summary(Boston[c("crim","tax","ptratio")])
```

crim	tax	ptratio
Min. : 0.00632	Min. :187.0	Min. :12.60
1st Qu.: 0.08205	1st Qu.:279.0	1st Qu.:17.40
Median : 0.25651	Median :330.0	Median :19.05
Mean : 3.61352	Mean :408.2	Mean :18.46
3rd Qu.: 3.67708	3rd Qu.:666.0	3rd Qu.:20.20
Max. :88.97620	Max. :711.0	Max. :22.00

From re-ranked tables and summary statistics we can see that:

- Crime is right-skewed with a median about 0.257 vs. a max of 88.976, min of 0.006. A small number of tracts have very high crime relative to the bulk.
- Tax also has a long upper tail: many of the highest-tax tracts sit at 666/711, which stand far above the median 330.
- Pupil-teacher ratio varies in a narrower region of 12.6 to 22.0. There are higher values, but nothing like the extreme outliers seen for crim and tax.

12) Describe the distribution of pupil-teacher ratio among the towns in this data set that have a per capita crime rate larger than 1. How does it differ from towns that have a per capita crime rate smaller than 1?

```
library(dplyr)
```

Attaching package: 'dplyr'

The following object is masked from 'package:MASS':

```
select
```

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(ggplot2)
```

Attaching package: 'ggplot2'

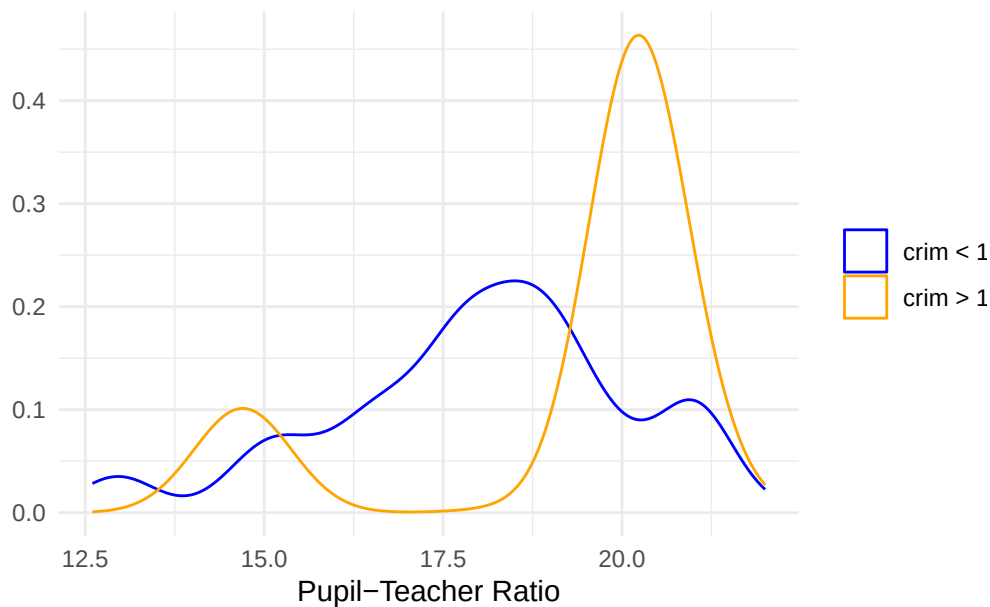
The following objects are masked from 'package:psych':

%+%, alpha

```
df <- Boston %>%
  filter(crim != 1) %>%
  mutate(group = if_else(crim > 1, "crim > 1", "crim < 1"))

ggplot(df, aes(x = ptratio, color = group)) +
  geom_density() +
  labs(title = "Pupil-teacher Ratio Distribution by Crime Rate",
       x = "Pupil-Teacher Ratio", y = " ") +
  scale_color_manual(values = c("crim > 1" = "orange", "crim < 1" = "blue")) +
  theme_minimal() +
  labs(color=NULL)
```

Pupil-teacher Ratio Distribution by Crime Rate



```
by(df$ptratio, df$group, summary)
```

```
df$group: crim < 1
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
12.60	16.80	18.30	18.02	19.20	22.00

```
-----
```

```
df$group: crim > 1
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
14.70	20.20	20.20	19.29	20.20	21.20

- From the summary and the density plot we can see that towns with $\text{crim} > 1$ have higher and tightly concentrated pupil-teacher ratios of median 20.20, mean 19.29, range from 14.70 to 21.20. The density is sharply peaked near 20.2 with a left tail.
- By contrast, towns with $\text{crim} < 1$ show lower and more dispersed ratios of median 18.30, mean 18.02, ranging from 12.60 to 22.00. Its density is flatter and more symmetric.

Writing Functions

13) Write a function that calculates 95% confidence intervals for a point estimate. The function should be called `my_CI`. When called with `my_CI(2, 0.2)`, the function should print out “The 95% CI upper bound of point estimate 2 with standard error 0.2 is 2.392. The lower bound is 1.608.”

```
my_CI <- function(pe, se, z = 1.96) {
  hi <- pe + z * se
  lo <- pe - z * se
  cat(paste0("The 95% CI upper bound of point estimate ", pe, " with standard error ", se, "
})

# test
my_CI(2, 0.2)
```

The 95% CI upper bound of point estimate 2 with standard error 0.2 is 2.392. The lower bound

*Note: The function should take a point estimate and its standard error as arguments. You may use the formula for 95% CI: point estimate +/- 1.96*standard error.*

Hint: Pasting text in R can be done with: paste() and paste0()

14) Create a new function called my_CI2 that does that same thing as the my_CI function but outputs a vector of length 2 with the lower and upper bound of the confidence interval instead of printing out the text. Use this to find the 95% confidence interval for a point estimate of 0 and standard error 0.4.

```
my_CI2 <- function(pe, se, z = 1.96) {
  half <- z * se
  c(lower = pe - half, upper = pe + half)
}

# test
my_CI2(0, 0.4)
```

```
lower upper
-0.784  0.784
```

15) Update the my_CI2 function to take any confidence level instead of only 95%. Call the new function my_CI3. You should add an argument to your function for confidence level.

```
my_CI3 <- function(pe, se, conf.level = 0.95) {
  z <- qnorm(1 - (1 - conf.level)/2)
  half <- z * se
  c(lower = pe - half, upper = pe + half)
}
```

```
# test
my_CI3(0, 0.4, 0.95)
```

```
      lower      upper
-0.7839856  0.7839856
```

```
my_CI3(0, 0.4, 0.99)
```

```
      lower      upper
-1.030332  1.030332
```

Hint: Use the `qnorm` function to find the appropriate z-value. For example, for a 95% confidence interval, using `qnorm(0.975)` gives approximately 1.96.

16) Without hardcoding any numbers in the code, find a 99% confidence interval for Ethnic Heterogeneity in the Angell dataset. Find the standard error by dividing the standard deviation by the square root of the sample size.

```
my_CI3(mean(angell_stata$ethhet),
      sd(angell_stata$ethhet) / sqrt(length(angell_stata$ethhet))
      , conf.level = 0.99)
```

```
      lower      upper
23.35425 39.38993
```

17) Write a function that you can `apply` to the Angell dataset to get 95% confidence intervals. The function should take one argument: a vector. Use if-else statements to output NA and avoid error messages if the column in the data frame is not numeric or logical.

```
CI95 <- function(v) {
  if (is.numeric(v) || is.logical(v)) {
    mu <- mean(v, na.rm = TRUE)
    se <- sd(v, na.rm = TRUE) / sqrt(sum(!is.na(v)))
    z <- qnorm(0.975)
    c(lower = mu - z * se, upper = mu + z * se)
  }
  else {
    cat("NA")
  }
}
```

```
# test
CI95(angell_stata$ethhet) # should output CI
```

```
      lower      upper
25.27127  37.47292
```

```
CI95(angell_stata$city) # should output NA
```

```
NA
```